

BISWADIP BISWANATH BHATTACHARYYA

Data Scientist

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SUMMARY

Data Scientist with around 3 years of experience driving predictive analytics and model deployment to improve retention, conversion, and operational efficiency. Skilled in Python, SQL, MLflow, MLOps, Docker, Kubernetes, XGBoost, time-series forecasting, and LLMs. Proficient in delivering measurable business impact through ML pipelines, A/B testing, and insight-driven dashboards while ensuring compliance and reliability.

SKILLS

Methodologies: SDLC, Agile, Waterfall

Programming Languages: Python, R, SQL

Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Keras, NLTK, XGBoost, LightGBM

Databases: Oracle, MySQL, MongoDB, SQL Server

Data Visualization: Tableau, Power BI, Excel (Advanced), Matplotlib, Seaborn

Cloud Platforms: AWS (EC2, S3, RDS, Glue, RedShift, SageMaker), Microsoft Azure (Data Lake, Data Factory, Synapse Analytics), Snowflake

Statistical Techniques: A/B Testing, Hypothesis Testing, Time Series Analysis (ARIMA)

Machine Learning Algorithms: Random Forest, Clustering (K-Means, DBSCAN), Decision Trees, Linear & Logistic Regression, SVM, Deep Learning (CNN, RNN, Transformers), Hyperparameter Tuning, Feature Engineering, Model Optimization

Generative AI & LLMs: OpenAI API, LangChain, Hugging Face Transformers, Retrieval-Augmented Generation (RAG), Prompt Engineering

Version Control & Containerization: Git, GitHub, Docker, Kubernetes

EDUCATION

Master of Science in Applied Data Science | Syracuse University, School of Information Studies, Syracuse, NY

May 2025

Bachelor of Engineering | University of Mumbai, India

May 2023

PROFESSIONAL EXPERIENCE

Data Scientist | PNC, New York

October 2024 - Current

- Implemented **MLflow-driven MLOps pipeline** using **Azure Databricks**, **Terraform** and **GitHub Actions** to retrain **XGBoost** churn models, improving customer retention by 17% and ensuring scalable cloud-based **machine learning** deployment.
- Developed and deployed **XGBoost** credit risk classification model via **FastAPI** microservice on **Azure Kubernetes Service**, processing 15K+ daily applications with 89% approval prediction accuracy, 0.92 AUC-ROC, and sub-second inference latency.
- Developed segmentation using **K-Means** and deployed **ML models** to **Azure cloud**, integrating **Power BI** dashboards, boosting regional marketing conversions 18% while aligning **machine learning** lifecycle with enterprise **MLOps** standards.
- Performed **A/B testing** using **Python** and **statistical analysis** to enhance mobile feature adoption by 20%, presenting insights via **Power BI** while embedding **AI** considerations into user behavior analytics.
- Built **Azure ML** batch automation using **ARIMA forecasting models**, achieving 2.8 RMSE and 92% forecast reliability, supporting staffing optimization through scalable **machine learning** operations.
- Experimented with **GPT-based LLM** and **RAG techniques** for internal knowledge query automation, supporting financial analytics review acceleration while maintaining compliance and demonstrating early **AI** integration capabilities.

Data Scientist | Vivma Software Inc, India

January 2022 - July 2023

- Identified \$500K uplift by analyzing 2M transactions using **SQL**, **Python** and **hypothesis testing**, applying **machine learning** principles for pricing optimization via statistically validated predictive analytics models.
- Engineered recommendation engine using **R**, collaborative filtering and **Decision Trees**, processed retail behavior data through automated pipelines, increasing conversion rate by 22%.
- Applied **ANOVA-based** feature selection using **Python Pandas** and **SciPy**, improving model accuracy 11%, while advancing maturity of machine learning readiness through structured data preprocessing workflows.
- Developed **Tableau** dashboards for inventory trends and customer lifetime value tracking, enabling 15% reduction in operational cost by providing data-driven decision insights.
- Performed **cohort analysis** and demand trend **forecasting** using **PostgreSQL ETL workflows**, improving reporting turnaround by 20% while adhering to traditional **machine learning** methodologies appropriate to pre-LLM enterprise landscape.

PROJECTS

Portfolio Optimization using Bidirectional CNN-LSTM Attention Mechanism: Designed CNN-LSTM attention models forecasting user behavior, achieving 5.87 MSE and outperforming SARIMA benchmarks. Enabled real time portfolio rebalancing using predictive models, boosting returns by 18% and significantly reducing portfolio drawdowns

Interactive AI Medical Consultant (Python, Streamlit, Gemini 1.5 Flash): Architected a conversational AI agent using Python, Lang Chain, Gemini API, and RAG, enabling accurate context specific responses Engineered multimodal intent analysis & diagnostic module, delivering insights and enhancing customer experience and decision making.

MLOps Pipeline for a Real-Time Prediction Service: Containerized & deployed a Scikit-learn prediction model using Docker on AWS, provisioning infrastructure via Terraform as Code. Built CI/CD pipeline with GitHub Actions, automating retraining, validation, and zero-downtime deployment of models

CERTIFICATIONS

• AWS AI Practitioner

• Oracle Cloud Infrastructure 2025 Certified Generative AI Professional